The Barbero–Immirzi Field as a Hyperbolic Axion–Dilaton: a Fixed Kinetic Mixing, Tests of a Torsion Dark-Energy Field, and Predictions Locked Before the Data

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Abstract

We test the Barbero–Immirzi (BI) parameter of loop quantum gravity, promoted to a dynamical pseudoscalar coupled to torsion (the "Twist"), as late-time dark energy, using Planck, ACT DR6, SPT-3G, DESI DR2 and three supernova samples, with every test rule fixed and hashed before its data were seen. A minimal thawing Twist ties ΛCDM on Pantheon+; its best preference (DES Y5, $\Delta\chi^2 = -6.2$) vanishes when that sample is split at $z = 0.1$ (the whole gain is a low/high-redshift calibration offset, as for w_0–w_a), and weakens on Union3 (-1.8). We then point out that the $\mathcal{N} = 1$ supergravity embedding of the BI field (Gates, Ketov & Yunes) is a dilaton–axion pair on a hyperbolic field space of curvature $-2/3$ ($\alpha = 1$), which fixes the axion's kinetic coupling to $e^{\lambda\chi}$ with $\lambda = -2\sqrt{2/3}$, a structure known to produce apparent phantom crossing. A two-field solver, validated against published attractors to 10^{-10} and implemented in AxiCLASS (cross-check 2×10^{-5}), shows the fixed geometry is weak mixing ($\xi \simeq 0.82$ versus $\gtrsim 370$ needed for the strong-mixing DESI benefit). The remaining route, the BI angle as a fraction f of the dark matter, is dark matter coupled to the dilaton with strength $0.816f$; it bends the w_0–w_a reading toward DESI, but existing coupling bounds ($f \lesssim 6$–9%) allow only $\Delta w_a \simeq -0.05$ to -0.09. We report solar-system, inflation (n_s tension 2.4σ) and early-energy checks, a closure caveat in the coupled AxiCLASS build, and forward predictions hash-locked before DESI DR3, Euclid, LiteBIRD and LISA data exist.

1 Introduction

DESI DR2 BAO, combined with the CMB and supernovae, prefers an evolving dark-energy equation of state that appears to cross $w = -1$ [1]. A canonical field cannot cross $w = -1$, but two-field systems with kinetic mixing can *appear* to cross in a w_0–w_a analysis while obeying the null energy condition [2, 3]. Loop quantum gravity contains a natural pseudoscalar: the Barbero–Immirzi parameter γ. Promoted to a field it couples to torsion through the Holst or Nieh–Yan terms [4, 5]. We ask two questions: does the minimal BI field fit the late-time data, and what does its own supergravity geometry imply? Every test below had its rule written, timestamped and hashed before computing; failures are reported at the same size as successes.

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2 The minimal Twist against the data

The late field is a thawing pseudoscalar with $V = m^2F^2(1-\cos\phi/F)$, fitted in AxiCLASS with Cobaya to Planck low-ℓ, ACT DR6 (with Planck high-ℓ cut and lensing), SPT-3G D1, DESI DR2 BAO and one supernova sample, on a build in which we found and fixed two AxiCLASS bugs that affect $n = 1$ fields (documented separately).

Test (locked rule)	Result	Verdict
Pantheon+	$\Delta\chi^2 = +0.09$ vs ΛCDM	tie
DES Y5	-6.23 (1.3σ after look-elsewhere)	weak hint
DES Y5 split at $z = 0.1$	$+0.53$ ($z > 0.1$), -0.29 ($z \le 0.1$), $r = -0.06$	calibration-sensitive
DES-Dovekie / Union3	-1.33 / -1.82	weakened
Without DESI (DES Y5)	-5.33	not DESI-driven
Against w_0–w_a	worse by 10–13	loses
$H(z)$ chronometers, $f\sigma_8$ (no refit)	$\lvert\Delta\chi^2\rvert < 0.5$	neutral

The DES Y5 preference exists only when nearby and distant supernovae share one absolute magnitude; neither half prefers the field, and w_0–w_a shows the same pattern ($r = -0.17$). This matches independent analyses that trace the DES Y5 excess to low-redshift calibration [15, 16].

3 The BI field's supergravity geometry

Gates, Ketov & Yunes [6] show that a dynamical BI parameter in $\mathcal{N} = 1$ supergravity is a chiral dilaton–axion superfield $\Phi = \phi + ia$, with $\Omega = e^{\Phi+\bar\Phi}(\Phi+\bar\Phi)$ and $K = -3\ln(-\Omega/3)$, $\phi < 0$. The linear part of K drops out of the metric:

$$K_{\Phi\bar\Phi} = \frac{3}{(\Phi+\bar\Phi)^2}, \qquad \mathcal{L}_{\rm kin} = -\tfrac{1}{2}(\partial\chi)^2 - \tfrac{1}{2}e^{\lambda\chi}(\partial\tilde a)^2, \qquad \boxed{\lambda = -2\sqrt{2/3}}, \tag{1}$$

with $|\phi| = e^{\sqrt{2/3}\chi}$, $\tilde a = \sqrt{3/2}\,a$: the hyperbolic plane of curvature $-2/3$, i.e. the $\alpha = 1$ case of $\lambda = -2\sqrt{2/(3\alpha)}$ [7]. Toomey et al. fit this coupling to Planck + DESI and find $\lambda = -1.3^{+1.3}_{-1.5}$ [2]; the fixed value lies inside, a consistency found after their fit.

Matter coupling. If matter couples to the supergravity Jordan frame, $A^2 = -3/\Omega$ and $\alpha_{\rm DEF} = (2/\sqrt{3})(|\phi| - \frac{1}{2})$: a general-relativity point at $\phi = -\frac{1}{2}$ that is a Damour–Nordtvedt attractor, but Cassini pins the dilaton within $|\Delta\chi| < 0.0072\,M_{\rm Pl}$ today, paralleling the Brax–Martin no-go theorem [8]. A rolling dilaton therefore needs matter blind to it, which is natural in scale-invariant Einstein–Cartan gravity [9, 10].

Simplest potential. A constant superpotential gives $V = \frac{81}{2}W^2(-\phi-1)e^{-6\phi}/\phi^2$, with a turning point exactly at $\phi = -\frac{1}{2}$ and its minimum at $\phi = -\frac{2}{3}$, both at negative energy; an uplift is required. The point $\phi = -\frac{1}{2}$ is a local *maximum* of this potential, so a light dilaton cannot rest at the GR point with constant W.

4 The Spiral Twist: solver, validation, implications

Model. Einstein frame, matter blind to the dilaton: $\mathcal{L} = -\frac{1}{2}(\partial\chi)^2 - \frac{1}{2}e^{\lambda\chi}(\partial\theta)^2 - V_0e^{-\kappa\chi} - m^2F^2(1-\cos\theta/F)$, with λ fixed as above.

The decay constant is not fixed by itself. The canonical axion is $\sqrt{3/2}\,a/|\phi|$, so a period P in a gives $F = \sqrt{3/2}\,P/(2\pi|\phi|)$: F depends on where the dilaton sits. With matter blind to the dilaton, $\chi \to \chi + c$, $F \to Fe^{-\lambda c/2}$, $\theta_i \to \theta_i e^{-\lambda c/2}$, $m \to me^{\lambda c/2}$, $V_0 \to V_0e^{\kappa c}$ is an

exact symmetry (checked numerically to 10^{-9}), so a value of F alone constrains nothing. Our $F = (4/\pi)\sqrt{3/2}\,M_{\rm Pl}$ follows from the Nieh–Yan index (Chandía–Zanelli: $\int N = 4\pi^2\ell^2\nu$, so the BI angle has period $\kappa/(\pi\ell^2) = 8$ at $\ell = \sqrt{G}$; derivation and assumptions in the release repository, `NIEH_YAN_PERIOD.md`) with the Einstein–Cartan kinetic coefficient 3/4 [5]. In the supergravity embedding that coefficient holds at $|\phi| = 1$, so F is fixed only if the dilaton sits there; this point is not an extremum of V, so something must hold it. The other distinguished points give $F = (4/\pi)\sqrt{6} = 3.12\,M_{\rm Pl}$ (GR point) or $2.34\,M_{\rm Pl}$ (minimum of the constant-W potential, where the dilaton must be heavy). A pinned dilaton cannot also roll, so either the dilaton rolls and F is free, or it is pinned and the late field is the axion alone with a discrete F.

Validation. Exact ΛCDM and single-field limits; true $w \geq -1$ always; no ghost (kinetic matrix positive, sound speeds 1). The late-time attractors of the published autonomous system of [3] ($(\lambda_p, \xi) = (2, 10), (2, 100), (8, 20)$) are reproduced to $\leq 5\times10^{-10}$. An attempt to reproduce the benchmark of [2] did not succeed (inputs not fully stated). In AxiCLASS, an exponential field plus dark matter coupled as $m_c \propto e^{\beta\phi}$ agrees with the solver to 3×10^{-8} ($\beta = 0$) and 2×10^{-5} ($\beta = 0.05$); with $\kappa = 0.05$, $\beta = 0$ the full pipeline reproduces the ΛCDM χ^2 within 0.08.

Closure caveat. For $\beta \neq 0$ the coupled build closes on its actual $H(0)$, which differs from the input h by up to 0.6%; reported H_0 and σ_8 labels carry this offset, χ^2 does not.

Weak mixing. In the notation of [3, 11], $\xi = -\lambda/2 = 0.82$ (or 1.63, depending on whether the kinetic function multiplies $(\partial a)^2$ once or squared), while their DESI benefit requires $\log_{10}\xi > 2.5$. A scan (with F fixed, which by the symmetry above is equivalent to free F over the scanned m, θ_i) gives at best $w_0 \simeq -0.84$, $w_a \simeq -0.24$, the thawing relation.

Matter-like angle. An oscillating angle has $m_{\rm eff} = m\,e^{-\lambda\chi/2}$: dark matter coupled to the dilaton with $d\ln m/d\chi = 0.816$, i.e. $|\beta_{\rm eff}| \simeq 0.816 f$ for a fraction f of the matter (negative in the convention $m \propto e^{-\beta\kappa\phi}$ of [14]). The dilaton rolls to larger χ, so this mass grows with time: the apparent-phantom branch of [14], consistent with the bend toward DESI below.

κ	f	$\beta_{\rm eff}$	w_0	w_a	thawing w_a
1.2	4.2%	0.034	−0.777	−0.356	−0.341
1.2	9.7%	0.079	−0.795	−0.401	−0.314
1.2	18.1%	0.148	−0.830	−0.599	−0.260

Coupled thawing quintessence bounds $|\beta| < 0.073$ [14] ($f \lesssim 9\%$), and our coupled fit prefers $|\beta| \simeq 0.05$ ($f \simeq 6\%$); the bend is then $\Delta w_a \simeq -0.05$ to -0.09. A full fit of this effective model (two dark-energy parameters, as for w_0–w_a) is running under a locked rule.

5 Further checks

Check	Result	Verdict
Inflation from $\alpha = 1$	$n_s = 0.9667$ at $N = 60$ vs 0.974 ± 0.003 [17]	tension (2.4σ)
Same, matching ACT's n_s	needs $N = 69$–87, then $r = 0.0016$–0.0025, at the locked kill line $r < 0.002$	squeezed
Decay constant F (Spiral)	exact degeneracy with the dilaton's value (above)	absorbed
Nieh–Yan F, late field (grid read)	DES Y5 $\Delta\chi^2 = -5.5$ vs ΛCDM, $+0.7$ vs best; Pantheon+ -0.4; Union3 not yet run	indicative
GW parity (Einstein–Cartan)	torsion integrates out to GR + scalar [5]: no tree-level birefringence	see §6
Dilaton as early dark energy	$\leq 2.5\%$ at $z = 3500$; G changes $\times 13$ at BBN	too small
Torsion-natural coupling $\beta = 0.816$ (all dark matter)	$\Delta\chi^2 = +71{,}655$	excluded
Minimal-field CMB birefringence	$\lvert\beta_{\rm rot}\rvert \approx 0$ vs 0.24°	excluded

6 Predictions locked before the data

Hash-locked on 2 November 2026 (SHA-256 begins `b629f534`; file and numbers archived with doi:10.5281/zenodo.23899237): DESI DR3 D_M/r_d, D_H/r_d at the DR2 redshifts (preferred if $\Delta\chi^2 \leq -4$); $w(z) \geq -1$ at all z for the minimal and coupled fields; S_8 unchanged (minimal) or 0.828 (coupled); CMB birefringence ratio $R = 0.997$ if photon-coupled; tensor ratio $r = 0.0033$–0.0048 for $\alpha = 1$ inflation (excluded if $r < 0.002$); gravitational-wave velocity birefringence $\sim 10^{-43}$–10^{-42} GeV. *Caveat found after locking:* that scale comes from teleparallel Nieh–Yan gravity [12]; in the Einstein–Cartan framework used here the Nieh–Yan-coupled field leaves GR plus a scalar [5], with no tree-level birefringence (a Holst-coupled field has been argued to give GW friction instead [13]). The locked null-result clause is unaffected.

7 The coupling: Nieh–Yan

A dynamical BI field can couple through the Holst or the Nieh–Yan term; after torsion is solved, Holst gives the canonical field $\sqrt{3/2}\, M_{\rm Pl} \operatorname{arsinh} \beta$ and Nieh–Yan the linear $\sqrt{3/2}\, M_{\rm Pl}\, \beta$ [5]. We adopt **Nieh–Yan**: the decay constant above is derived from its index; it is shift-symmetric, as the GKY axion is (the Kähler potential depends only on $\Phi + \bar{\Phi}$); and every fit already uses the cosine in the canonical displacement, which is its form. The lopsided "Fingerprint" potential is then the Holst alternative, and its pre-registered comparison with the plain cosine becomes a data test of the coupling. The BI rest point $\gamma_0 = 0.2375$ has no observable content under the Nieh–Yan shift symmetry. Decisive next tests: the locked fixed-F fit on Union3 (blind) and the Fingerprint fit.

8 Conclusions

The minimal BI dark-energy field is consistent with the data but not preferred; its strongest hint is a supernova calibration effect. The BI field's own supergravity geometry fixes a kinetic mixing of the type behind apparent phantom crossing, but too weakly to matter unless part of the dark matter is the BI angle, which existing bounds keep small. The locked predictions above will decide the remaining routes.

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